

Technical Addendum: Ordered Relative Field Receipts for Pairwise Multiscale Market Comparison

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Abstract

This addendum specifies Relative Field Pair v1, a versioned comparison receipt over two independently constructed 15-coordinate Market Field instances. The operator enforces a common component recipe, timeframe-specific exact shared-key intersection, bilateral coordinate support, independent proper-fit normalization, full-precision relative-price context, a strictly prior rolling beta adjustment, ordered provenance, and zero decision authority. We give the alignment key, availability masks, native and fit-relative differences, relative-index and beta-adjusted-return equations, family-balanced relationship-stretch rule, precision boundaries, receipt hash, algebraic invariants, failure modes, and test map needed to reconstruct the implementation. The method does not select peers, estimate cointegration, infer connectedness, identify a winner, or establish predictive or economic value. Its contribution is an auditable comparison and translation layer, not a relative-value trading model.

1 Technical Summary

Relative Field Pair v1 answers a deliberately narrow question: given two instruments analyzed by the same Market Field recipe, what can be compared without hiding timestamp mismatch, unsupported coordinates, self-history normalization, or downstream authority? It produces three parallel descriptive channels:

1. a normalized relative-price path;
2. a prior-only beta-adjusted log-return path; and
3. native-scale and own-history-relative gaps for each of the 15 field coordinates.

The price channels do not transform the field coordinates. The coordinate channels do not alter either component dictionary, match cross-symbol Form identifiers, or supply a trade score.

Supported claim. Pair v1 integrates exact alignment, bilateral support, independent fixed-reference normalization, ordered identity, and explicit zero authority in one reconstructible receipt. Unsupported claims. It is not evidence of alpha, mean reversion, market neutrality, factor neutrality, leadership, spillovers, causal propagation, prediction, option efficacy, or profitability.

2 Relationship to Established Methods

Benchmark-relative returns and market-model slopes are established tools. Brown and Warner study event-window abnormal-return methods using estimated return models (Brown & Warner, 1985). Pair v1 borrows only the familiar covariance/variance slope form. It defines no event, estimates no event-window effect, performs no cross-sectional inference, and does not subtract a fitted intercept from its displayed beta-adjusted return.

Displayed object	Exact role	Not implied
Relative index	Ratio of two normalized full-precision close paths	Cointegration, fair value, or convergence
Beta-adjusted path	Target log return less a prior-window slope times benchmark log return	OLS alpha, market neutrality, or abnormal-return inference
Native gap	Target minus benchmark on one implemented coordinate scale	Raw market-unit difference or utility ordering
Context gap	Difference between two own-fit standardized coordinates	Cross-sectional z-score or persistent common state
Relationship stretch	Family-balanced absolute context-gap change over five aligned observations	Leadership, lead-lag, cycle, or connectedness

Table 1: The three comparison channels are intentionally distinct.

Likewise, normalized price distance and convergence rules are central to classical pairs-trading research (Gatev et al., 2006). Pair v1 has no pair-formation optimization, trading band, entry or exit rule, convergence assumption, transaction-cost model, or return claim. It also performs no unit-root, residual-stationarity, cointegrating-vector, or error-correction test of the type associated with cointegration analysis (Engle & Granger, 1987). Finally, the coordinate gaps are not a variance-decomposition network: no VAR or forecast-error variance decomposition is estimated, so the output is not Diebold–Yilmaz connectedness (Diebold & Yilmaz, 2012).

The implementation contribution is therefore narrower: co-register two independently calculated multiscale fields, expose where both are actually supported, compare each instrument both natively and relative to its own fixed fit history, keep price context separate, and bind the result to an ordered receipt.

3 Inputs, Coordinate Space, and Component Chronology

3.1 Independent component fields

Let $q \in \{A, B\}$ denote the target and benchmark. Each leg begins with a normalized OHLCV history and is independently mapped by a component recipe $\mathcal{R}$ to

$$\{X_{j,t}^q, M_{j,t}^q\}_{j=1}^{15}, \quad m_j^q, s_j^q, e_q, H_q, \quad (1)$$

where $X_{j,t}^q$ is coordinate j , $M_{j,t}^q$ is its full-dependency-support indicator, m_j^q and s_j^q are the stored proper-fit location and scale, e_q is the evaluation start, and H_q is the component analysis hash. The service requires equal recipe hashes. That equality binds the formula version, semantic revision, ordered horizon grid, settings, normalization contract, EWM convention, and retrospective-research inclusion flag. It does not assert equal histories or equal providers.

For requested visible length V and largest horizon $h_{\max}$, each component asks the history layer for

$$V_{\text{fetch}} = V + \max(72, 2h_{\max}) \quad (2)$$

rows and requires at least

$$n_{\min} = \max(60, h_{\max} + 1) \quad (3)$$

returned rows. The field and its proper-fit/calibration/evaluation split are calculated on that retained component history. Pair output is subsequently restricted to the final V common aligned observations. Consequently, earlier component rows may initialize the field and its dictionary but are not used by the visible-window relative-price base or beta estimator.

3.2 The ordered 15-coordinate space

The comparison preserves the component coordinate order and family labels in Table 2. “Native” always means the coordinate’s implemented model scale, not raw market units.

j	Coordinate	Family	Implemented unit / polarity
1	Pressure	pressure state	bounded signed
2	Pressure change	pressure state	bounded signed
3	Acceleration	pressure state	bounded signed
4	Jerk	pressure state	bounded signed
5	Snap	pressure state	bounded signed
6	Structure	field transform	$[0, 1]$ descriptive score
7	Kinematics	field transform	$[0, 1]$ descriptive score
8	Geometry	field transform	$[0, 1]$ descriptive score
9	Information	field transform	$[0, 1]$ descriptive score
10	Propagation	field transform	$[0, 1]$ descriptive score
11	Cascade bias	field transform	bounded signed
12	Scaling exponent	field transform	log-horizon slope, descriptive
13	Volatility vs baseline	OHLCV carrier	$[0, 1]$ causal-baseline level
14	Participation vs baseline	OHLCV carrier	$[0, 1]$ causal-baseline level
15	Liquidity stress vs baseline	OHLCV carrier	$[0, 1]$ level; lower is less stressed

Table 2: Exact Pair-v1 coordinate order. Higher is not inherently better.

The three carrier values are bounded transformations relative to causal baselines; they are not raw realized variation, volume, or price-impact units. Likewise, larger disorder, propagation, participation, or liquidity stress is not a favorable ranking.

4 Exact Alignment and Availability Contract

4.1 Alignment keys

Let s be a serialized component timestamp and τ the requested timeframe. Pair v1 constructs

$$\kappa_\tau(s) = \begin{cases} \text{date}(s), & \tau \in \{1D, 1W\}, \\ \text{ISO}(\text{UTC}(s)), & \tau \notin \{1D, 1W\}, s \text{ timezone-aware}, \\ \text{ISO}(s), & \tau \notin \{1D, 1W\}, s \text{ timezone-naive}. \end{cases} \quad (4)$$

If K_q is leg q ’s valid key set, define

$$K_{AB}^{\text{all}} = \text{sort}(K_A \cap K_B), \quad (5)$$

$$K_{AB} = \text{tail}_V(K_{AB}^{\text{all}}). \quad (6)$$

The explicit window counts are

$$N_{\text{requested}} = V, \quad (7)$$

$$N_{\text{available}} = |K_{AB}^{\text{all}}|, \quad (8)$$

$$N_{\text{returned}} = |K_{AB}| = \min(V, N_{\text{available}}). \quad (9)$$

The response also returns $|K_A|$, $|K_B|$, the first and last returned keys, and whether $N_{\text{available}} > N_{\text{returned}}$. These counts separate the requested display window from what exact alignment made available. At least 20 exact shared observations must exist before the visible tail is selected. No nearest-neighbor match, interpolation, forward fill, or carry-forward is permitted.

Daily and weekly dates are provider-serialized date labels, not an independent exchange-calendar certification. Aware intraday times are normalized to UTC before exact equality; naive intraday times remain naive and require exact serialized equality. A matching key does not prove the two observations describe the same economic session. Session compatibility is reported as compatible only for the same canonical symbol and otherwise remains unknown.

4.2 Visible-window dropout accounting

Let $k_0 = \min K_{AB}$ and $k_T = \max K_{AB}$. For leg q , the displayed dropped count is

$$D_q = \# \{k \in K_q : k \geq k_0, k \notin K_{AB}\}, \quad (10)$$

and the unmatched tail count is

$$U_q = \# \{k \in K_q : k > k_T\}. \quad (11)$$

Thus U_q is a subset of D_q and must not be added to it. These quantities do not count pre-window truncation or rows rejected before the pair layer during OHLCV normalization. The receipt reports counts, not the complete list of dropped timestamps.

4.3 Coordinate support

For each leg, $M_{j,t}^q = 1$ only when the coordinate has reached its declared first full-dependency-support timestamp and the serialized value is finite. Scaling exponent additionally requires its validity flag. Each carrier additionally requires a finite corresponding causal-ratio observation. Define bilateral support

$$S_{j,t} = M_{j,t}^A M_{j,t}^B \mathbf{1}[X_{j,t}^A, X_{j,t}^B \text{ finite}]. \quad (12)$$

The support fraction in the response is

$$\text{SupportFraction} = \frac{C_{\text{supported}}}{C_{\text{total}}}, \quad C_{\text{supported}} = \sum_{t \in K_{AB}} \sum_{j=1}^{15} S_{j,t}, \quad C_{\text{total}} = 15N_{\text{returned}}. \quad (13)$$

The response reports both cell counts, their ratio, the literal bilateral full-dependency-support rule, whether every returned cell is supported, and `missing_values_carried=false`. This is visible-window native-coordinate support. It is neither latest-coordinate support, context support, nor a price-overlap fraction.

4.4 DXY routing boundary

Requested symbols DXY, ^DXY, and DX-Y.NYB canonicalize to DXY and retain Yahoo's provider symbol DX-Y.NYB; the system does not silently substitute the tradable ETF UUP. Because the current provider bar anchors are not demonstrated compatible, nonidentity DXY comparisons at 1h, 2h, and 4h are rejected before provider access. DXY/DXY remains available as an identity control. Permitting another timeframe is not a claim that its sessions are economically synchronized.

5 Coordinate Comparison Operators

5.1 Native difference

The native operator is

$$\Delta_{j,t}^N = \begin{cases} X_{j,t}^A - X_{j,t}^B, & S_{j,t} = 1, \\ \text{NA}, & \text{otherwise.} \end{cases} \quad (14)$$

An unavailable cell remains unavailable; it is never interpreted as a zero difference.

5.2 Own-history fit-relative difference

For completeness, the component dictionary obtains m_j^q from the coordinate median on its proper-fit segment. Its robust scale is selected deterministically:

$$s_j^q = \begin{cases} Q_{0.75} - Q_{0.25}, & Q_{0.75} - Q_{0.25} > 10^{-9}, \\ 1.4826 \text{ median } |X - m_j^q|, & \text{if that value} > 10^{-9}, \\ \text{Std}_{\text{pop}}(X), & \text{if that value} > 10^{-9}, \\ 1, & \text{otherwise.} \end{cases} \quad (15)$$

All quantities in Equation 15 use that leg’s proper-fit rows. $Q_{0.25}$ and $Q_{0.75}$ use NumPy method=linear interpolation. Pair v1 consumes the stored fit references rather than recomputing them. A stored scale must be finite and greater than 10^{-12} .

Define

$$Z_{j,t}^q = \frac{X_{j,t}^q - m_j^q}{s_j^q}. \quad (16)$$

The context difference is

$$\Delta_{j,t}^C = \begin{cases} Z_{j,t}^A - Z_{j,t}^B, & S_{j,t} = 1, \ t \geq \max(e_A, e_B), \\ \text{NA}, & \text{otherwise.} \end{cases} \quad (17)$$

It compares how unusual the two coordinates are relative to their separate proper-fit histories. It is not a cross-sectional economic z-score. Fit and held-out calibration rows are not backfilled with later reference statistics, and request-local Form identifiers or centroids are never matched between symbols.

5.3 Precision boundary

Pair-v1 field gaps operate on the public component receipt. Coordinate series values enter from the component’s four-decimal serialization; fit medians and robust scales enter from its six-decimal lexicon serialization. Each $Z_{j,t}^q$ is serialized to six decimals before $\Delta_{j,t}^C$ is formed, and returned gaps are serialized to six decimals. These conventions make the receipt reconstructible from the component payloads but mean that the coordinate calculus is not a full-precision recomputation.

Price comparison follows a separate path. The service retains full-precision normalized closes privately beside each component analysis, computes relative price and beta adjustment from those closes, and only then serializes returned closes to eight decimals and return quantities to six decimals.

5.4 Chronology and request-window dependence

For a fixed request, native values are prefix-derived and context begins only after both legs’ fixed proper-fit and calibration segments. Yet the context trace is not invariant to changing the request history: a different retained window can move split locations and change m_j^q or s_j^q . Likewise, changing V changes the first visible common close and therefore rebases the relative index. These are request-relative diagnostics, not persistent cross-symbol coordinates.

6 Relative Price and Prior-Only Beta Adjustment

6.1 Normalized relative price

Index the visible common keys as $t_0 < \dots < t_{n-1}$ and let $p_i^q > 0$ be the full-precision close of leg q at t_i . The relative index and active progress are

$$RI_i = 100 \frac{p_i^A / p_0^A}{p_i^B / p_0^B}, \quad (18)$$

$$AP_i = RI_i - 100. \quad (19)$$

The index is a rebased price ratio. It does not control currency, dividends, corporate actions, or provider adjustment policy beyond what is present in the returned component histories.

6.2 Strictly prior slope

Define aligned log returns

$$r_i^q = \log \frac{p_i^q}{p_{i-1}^q}, \quad i = 1, \dots, n-1. \quad (20)$$

For current return i , the strictly prior estimation set is

$$\mathcal{W}_i = \{\max(1, i - 60), \dots, i - 1\}, \quad N_i = |\mathcal{W}_i|. \quad (21)$$

No estimate is available unless $N_i \geq 20$. With

$$\bar{r}_i^q = \frac{1}{N_i} \sum_{u \in \mathcal{W}_i} r_u^q, \quad (22)$$

define

$$V_i^B = \sum_{u \in \mathcal{W}_i} (r_u^B - \bar{r}_i^B)^2, \quad (23)$$

$$\sigma_i^B = \sqrt{\frac{V_i^B}{N_i}}, \quad (24)$$

$$\hat{\beta}_i^- = \frac{\sum_{u \in \mathcal{W}_i} (r_u^A - \bar{r}_i^A)(r_u^B - \bar{r}_i^B)}{V_i^B}. \quad (25)$$

This is the centered covariance/variance slope, equivalent to the slope from an intercept-inclusive OLS fit. The fitted intercept is neither returned nor subtracted below.

The estimate is unavailable if $N_i < 20$, the two prior arrays differ in length, $\sigma_i^B < 10^{-7}$, $\hat{\beta}_i^-$ is nonfinite, or $|\hat{\beta}_i^-| > 25$. Equality at the two numerical thresholds is accepted. Estimates outside the gate are rejected, not clipped.

6.3 Beta-adjusted chain

The current increment is

$$e_i = r_i^A - \hat{\beta}_i^- r_i^B. \quad (26)$$

Because Equation 26 omits a fitted intercept, e_i is a beta-adjusted log-return differential, not an OLS residual, abnormal return, or alpha. Let C_i be the cumulative log differential:

$$C_i = \begin{cases} \text{NA}, & \hat{\beta}_i^- \text{ unavailable,} \\ e_i, & \hat{\beta}_i^- \text{ available and } C_{i-1} \text{ unavailable,} \\ C_{i-1} + e_i, & \text{otherwise.} \end{cases} \quad (27)$$

The displayed cumulative value is

$$BA_i = 100\{\exp(C_i) - 1\}. \quad (28)$$

An unavailable beta resets the chain. A later valid beta starts a new contiguous chain, and the latest summary never carries a stale estimate. Every price row reports the number of prior return pairs, nullable chain identifier, chain-start marker, and reset marker. The current summary separately reports the configured and realized prior-return counts, current-chain start and end, current-chain observation count, total chain and reset counts, and last-reset timestamp. Because the estimator is built only from visible common returns, its first possible value requires 20 prior returns plus the current return—22 aligned prices including the base row. If common keys contain time gaps, the corresponding log returns span those gaps; no elapsed-time normalization is applied.

7 Family-Balanced Relationship Stretch

Let t be the latest common key and t^{-5} the key five common observations earlier. For family $f \in \{\text{pressure state, field transform, OHLCV carrier}\}$, define the common endpoint support

$$J_f = \left\{ j : F(j) = f, \Delta_{j,t}^C \text{ and } \Delta_{j,t^{-5}}^C \text{ finite} \right\}. \quad (29)$$

Using the same J_f at both endpoints prevents support appearing or disappearing from masquerading as relationship movement. For $u \in \{t, t^{-5}\}$,

$$G_{f,u} = \frac{1}{|J_f|} \sum_{j \in J_f} |\Delta_{j,u}^C|, \quad (30)$$

$$S_u = \frac{1}{3} \sum_f G_{f,u}. \quad (31)$$

The summary is unavailable if any J_f is empty. With

$$\eta = \max(0.05, 0.05S_{t^{-5}}), \quad (32)$$

the label is

$$\text{StretchLabel} = \begin{cases} \text{widening,} & S_t > S_{t^{-5}} + \eta, \\ \text{converging,} & S_t < S_{t^{-5}} - \eta, \\ \text{mixed,} & \text{otherwise.} \end{cases} \quad (33)$$

The three families receive equal total weight despite containing 5, 7, and 3 coordinates. The response returns S_t , $S_{t^{-5}}$, their change, η , the lookback, and the counts of participating families and coordinates alongside the categorical value. The reader-facing translation maps mixed to “No clear net change” because the branch includes movement that remains within tolerance; it is not evidence of a statistical mixture. Stretch measures standardized separation magnitude, not direction or leadership.

8 Algebraic and Software Invariants

Identity control. If the canonical symbol and component analysis are identical, then every supported native and context gap is zero and $RI_i = 100$. Where the beta gate is admissible and benchmark variance is positive, $\hat{\beta}_i^- = 1$ and the beta-adjusted increment is zero; a locally constant return window remains unavailable rather than being forced to zero. The receipt labels the pair as an identity control.

Swap behavior. For a fixed common key set and fixed component references,

$$\Delta_{AB,j,t}^N = -\Delta_{BA,j,t}^N, \quad \Delta_{AB,j,t}^C = -\Delta_{BA,j,t}^C. \quad (34)$$

Before response rounding,

$$RI_{BA,i} = \frac{10000}{RI_{AB,i}}. \quad (35)$$

Thus active progress does not simply change sign. The rolling beta-adjusted paths are directional estimators and need not be antisymmetric. Absolute relationship stretch and bilateral support are swap-invariant.

Missingness preservation. Equations 14 and 17 never zero-fill an unsupported coordinate. The context gap is unavailable before the shared evaluation boundary, and Equation 27 never carries a beta-adjusted chain across an invalid estimate.

Prior-only beta property. At row i , Equation 25 uses returns only through $i - 1$. Mutating prices after any earlier row cannot change that row’s beta or cumulative beta-adjusted path. This is a computational prefix property, not causal inference.

Ordered identity. Swapping target and benchmark changes the ordered receipt preimage unless the two leg identities coincide. A cryptographic digest is used as an identity receipt, not as proof that the provider supplied correct, immutable, or exchange-complete bars.

9 Receipt, Cache, and Authority Contracts

9.1 Component and ordered comparison hashes

Each component analysis hash has the form

$$H_q^{\text{recipe}} = \text{SHA256}(\text{canonical recipe payload}), \quad (36)$$

$$H_q^{\text{input}} = \text{SHA256}(\text{normalized OHLCV rows}), \quad (37)$$

$$H_q = \text{SHA256}(\{H_q^{\text{recipe}}, H_q^{\text{input}}\}). \quad (38)$$

The ordered comparison preimage contains:

- Pair schema `market_field_pair_v1`;
- ordered target canonical symbol, provider symbol, and H_A ;
- ordered benchmark canonical symbol, provider symbol, and H_B ;
- timeframe;
- alignment contract `pair_alignment_v1`;
- normalization contract `native_and_fixed_proper_fit_relative_v1`; and
- $\text{SHA256}(K_{AB})$ for the ordered visible common keys.

Canonical JSON uses sorted keys, compact separators, finite JSON numbers, ASCII escaping, UTF-8 bytes, canonical positive zero, and SHA-256. Requested aliases such as DXY versus $\hat{\text{DXY}}$ remain visible provenance but are not distinct comparison-hash fields after they resolve to the same canonical and provider symbols. Provider name and data-source labels are also not committed separately. The API cache key includes the requested aliases so an alias-specific provenance receipt is not served from a differently requested alias.

9.2 Frozen result receipt and deterministic translation

The comparison hash identifies the ordered calculation contract; a separate `market_field_pair_receipt_v1` freezes the returned evidence needed to preserve one observed result. Let R^- denote a canonical JSON object containing:

- the Pair and semantic schema identifiers, timeframe, and shared-window endpoint;
- requested, canonical, and provider identities for both legs, their analysis hashes, data-source labels, and latest aligned/returned closes;
- the requested, available, and returned exact shared-window counts, per-leg dropped counts, unmatched tails, and latest-returned timestamps;
- the complete ordered shared-key sequence, $\text{SHA256}(K_{AB})$, and alignment contract;
- coordinate-cell support counts and the explicit no-carry rule;
- session, currency, adjustment, and timestamp-alignment disclosures;
- the latest relative-index, beta-chain, field-separation, and 15-coordinate values;
- the component, comparison, recipe, alignment, and normalization identities; and
- the identity-control flag and zero-authority contract.

The service returns

$$R = R^- \cup \{\text{receipt_hash} : \text{SHA256}(R^-)\}. \quad (39)$$

The response header repeats this digest as X-Market-Weather-Receipt-Hash. Recomputing Equation 39 detects mutation of the serialized receipt body; it does not certify provider truth, bar completeness, economic validity, or independent custody. SHA-256 is used here as an unkeyed checksum, not a digital signature or proof of origin.

A separate `pair_summary_v1` is a deterministic descriptive translation of authoritative relative-progress, current beta-chain, coordinate-gap, field-separation, support, and session

fields. It introduces no fitted score, peer choice, forecast, or authority. Its three notable coordinate rows are

$$\mathcal{T}_t = \text{first}_3 \text{ sort}_{j: S_{j,t}=1, \Delta_{j,t}^C \text{ finite}} \left(-|\Delta_{j,t}^C|, j \right), \quad (40)$$

where recipe position j is the deterministic tie-breaker. Summary prose uses two-decimal display formatting while the numeric response retains its declared serialization precision. The live URL preserves the selector recipe and reruns with then-current data; exporting R preserves the current evidence boundary. Version 1 of R does not bind the summary wording, generation/cache metadata, caveat prose, full price path, or full coordinate histories and is therefore not a complete chart-replay archive.

9.3 Reader-facing evidence contract

The interface labels the construction an Auditable field recipe, not an independently audited one. Its Overview reports the all-window ratio in Equation 13 as supported coordinate cells over total coordinate cells and reports current bilateral coordinate support separately. It also separates observed session status from independent session certification. The normalized price panel names the exact common base date on which both price indexes equal 100.

Comparator selectors show a descriptive reference category—broad-cap, growth/technology-heavy, equal-weight, small-cap, dollar-index, sector, or custom—together with “User selected; suitability not evaluated.” They do not recommend or score a peer. The export control is labeled Export compact receipt · JSON; its pre-export description states that it preserves aligned keys, current measurements, support, identities, disclosures, and authority, excludes full chart histories, and is not digitally signed.

The reader-facing order is Overview, Field detail, and Audit receipt. Field detail mounts one of four relationship scopes at a time and exposes all 15 coordinate histories on demand; audit groups remain collapsed until requested. Visual identity is redundant rather than color-only: target, benchmark, and difference use solid/circle, dashed/diamond, and dotted/square encodings, respectively. Current scope points also differ in outline and size. Trail opacity follows source chronology rather than screen position, while bounded display simplification preserves supported endpoints and turns. Unsupported intervals remain broken or hatched. Synchronized inspection uses shared dates, and distinct start, first-supported, current, and latest-supported labels appear only when their conditions hold. Coordinate strips place signed microbars above or below a centerline, encode magnitude through height and intensity, and retain a latest-direction arrow. Beta-chain starts and restarts remain visible as discontinuities and labeled markers. The default inspection scale may zoom the selected trail; a linkable shared scale preserves subject comparability. These are translation conventions and do not change any Pair coordinate, support mask, or identity.

9.4 Operational caching

Raw histories use the persistent market-history cache. Pair calculations use a per-worker TTL/LRU analysis cache with single-flight coalescing and isolated response copies. The ordered symbols, provider object identities, timeframe, visible length, horizons, and settings participate in that operational key. Cache metadata reports whether the pair computation was a hit, miss, or coalesced wait and whether either leg called a provider on that request.

Cache status is not scientific evidence. A hit does not certify data freshness, session comparability, provider truth, or bar completion. The response generated_at records calculation time and may predate a cache-hit request.

The request-local market_field_pair_runtime_v1 object and Server-Timing header separate cache lookup/wait/copy work from a miss-owner build. On a build, measured stages distinguish each leg’s history access, field construction, price-row preparation, and pair assembly. Identity controls disclose benchmark-leg reuse rather than inventing a second-leg duration. The endpoint returns one JSON response, so the measured handler-to-response-ready boundary excludes FastAPI JSON serialization, middleware compression, network transfer, frontend rendering, and chart interaction. Those values remain explicitly unavail-

able rather than being filled with synthetic progressive stages. Runtime metadata is operational and request-specific; it is excluded from both the ordered comparison hash and Equation 39.

The browser separately emits a neutral `pair_surface_second_frame` diagnostic from component mount through two animation-frame callbacks. That quantity is not labeled as paint, first useful summary, or chart interactivity. It does not enter either analytical identity.

9.5 Zero authority

The response declares:

Field	Pair-v1 value
<code>mode</code>	<code>research_display_only</code>
<code>scanner_weight</code>	0.0
<code>option_learning_weight</code>	0.0
<code>veto</code>	false
<code>sizing</code>	false
<code>execution</code>	false

Pair data are consumed by the Market Field research API and page. Pair v1 does not enter option eligibility, scanner ranking, the separately governed outcome-learning canary, manager verdicts, targets, quantities, or execution. Human readers may still be behaviorally influenced by a displayed comparison; “zero authority” describes the application control path, not zero human exposure.

10 Verification Evidence

The focused backend suite uses deterministic synthetic OHLCV fixtures and endpoint doubles. At the current validation boundary, 37 focused Pair/cache/probe tests passed; the complete backend suite passed 396 tests with two skips. The suite establishes implementation properties, not market efficacy:

The complete frontend suite passed 146 tests. Its comparison, summary, methodology, page, query-state, and product-telemetry cases cover deterministic translation, Overview/Field/Audit hierarchy, selectors, URL round trips, DXY guards, loading/error behavior, clearing stale results, and bounded event payloads. The committed browser release gate passed 11 of 11 checks. It retains review screenshots at 1440, 1024, 768, and 390 pixels; checks horizontal overflow and escaped controls; exercises tab-key navigation, the mobile-dialog focus trap, Escape and focus restoration; runs axe scans configured to fail serious or critical WCAG A/AA findings over all three tabs and the mobile dialog; tests reduced-motion behavior; and exercises a deterministic 720-pixel reflow proxy for 200% zoom. A committed machine-readable probe validates cross-section schema invariants, 15 coordinates, alignment/support denominators, zero authority, and receipt header/body identity. Its opt-in production mode emits a JSON result that should be retained after deployment; the committed release gate does not mislabel a local or predeployment probe as postdeployment evidence.

These checks have narrower meanings than their shorthand names. Retained screenshots are review evidence, not pixel-diff baselines. Viewport emulation is not a real-device audit, the reflow proxy is not literal browser zoom, and axe does not replace a manual screen-reader or color-vision review. The live probe does not certify provider truth or a stable latency distribution. Backend response-ready timing is not time to first useful summary or chart interactivity.

Test area	Property exercised
Identity and schema	15 coordinates; literal Pair-v1 schema; zero supported gaps; $RI = 100$; identity receipt
Ordered swap	Signed native/context reversal and distinct ordered comparison identity
Relative-price reciprocity	$RI_{AB}RI_{BA} = 10,000$ within serialization tolerance; active progress is not treated as additive antisymmetry
Recipe guard	Unequal component recipe hashes are rejected
Chronology and support	Context withheld before both evaluation starts; unsupported cells remain unavailable
Alignment modes	Daily date alignment, exact aware-UTC intraday equality, and exact naive timestamp equality
Unmatched tails	Latest aligned and latest returned observations remain distinct; tail counts are exposed
DXY contract	Canonical alias, Yahoo provider symbol, hourly rejection, and identity exception
Prior-only beta	Future price mutation leaves earlier beta and cumulative values unchanged
Known-vector beta	Centered slope, sample/gate boundaries, and deliberate retention of the fitted intercept in the displayed differential
Invalid current beta	No stale beta or cumulative value is carried into the current summary
Beta-chain metadata	Interruption, reset, and recovery produce explicit chain identifiers, starts, counts, and dates
Input depth	Largest horizon increases the required leg depth
Price precision	Positive micro-price relative progress survives four-decimal component display rounding
Endpoint/cache	Identity uses one fetch and subsequent pair computation reuses the analysis cache
Drop/hash boundaries	Pre-window rows are excluded from dropout; tails are subsets; committed and uncommitted receipt fields are distinguished
Frozen evidence	Exact window/support disclosures, deterministic summary inputs, and canonical frozen-receipt self-hash replay
Canonical serialization	Signed zero normalizes before summary formatting and SHA-256 serialization

Table 3: Pair-specific verification scope. Several rows summarize multiple tests.

HTTP class	Pair endpoint behavior
400	Invalid ticker characters, timeframe, horizon order/count, or request-size bounds
422	Unsupported DXY alignment, recipe mismatch, fewer than 20 exact shared observations, or invalid aligned positive-price requirements
502	Provider or history-acquisition failure not resolved by the cache layer

Table 4: Public failure boundary. Unavailable coordinates and beta rows remain successful responses with explicit nulls.

11 Failure Semantics

The service does not independently certify exchange calendars, currency equivalence, adjustment-policy equivalence, corporate-action treatment, or completion of the latest bar. Matched alignment keys may still represent asynchronous or stale markets. Missing common observations produce irregular elapsed intervals in the relative-price and beta paths.

12 Limitations and Robustness Risks

1. No efficacy result. There is no prospective predictive, decision-support, or cost-aware economic evaluation for Pair v1.

2. User-selected comparators. Competitor, sector, SPY, QQQ, IWM, RSP, and DXY presets are convenience references. They are not learned, point-in-time validated, or guaranteed economically appropriate benchmarks.
3. Window dependence. Relative-index basing, component fit/calibration splits, and own-fit context can change with requested history.
4. Market harmonization. Currency, dividends, corporate actions, calendars, time-zones, market hours, and stale-price risk are not jointly normalized.
5. Beta fragility. A 20–60 observation slope can be outlier-sensitive and nonstationary. The standard-deviation and absolute-beta gates are engineering safeguards, not statistically selected thresholds. There are no uncertainty intervals.
6. Multiple views. Fifteen coordinates, many possible pairs, two gap bases, many windows, and several scope projections create substantial multiple-comparison and researcher-degree-of-freedom risk.
7. No persistent shared Forms. Each component dictionary is request-local. Cross-symbol Form identity, centroid matching, or transition comparison is intentionally absent.
8. Not multivariate. Pair v1 has no basket, cross-sectional rank, factor decomposition, covariance shrinkage, or connectedness network.
9. Visual semantics. Oscilloscope-like scope paths are two-dimensional projections. Loops are not detected cycles, attractors, lead-lag relations, or causal mechanisms.
10. Operational scale. Large visible windows serialize 15 complete coordinate paths plus price context. Cache policy and response compression affect latency but do not change the mathematics.
11. Human-exposure evidence. Bounded browser events distinguish a rendered result, a substantially visible Overview, opened Field/Audit panels, changed basis/coordinate, copied summary/live recipe, exported receipt, and same-session navigation to Secret Options. They are ordinary product telemetry, not an authenticated durable exposure ledger, and historical exposure cannot be reconstructed. The separate second-frame diagnostic is not evidence that the Overview was seen or useful.

13 Predeclared Validation and Extension Plan

The present implementation is suitable for prospective instrumentation, not promotion into option ranking. A defensible next study should freeze the following before reading outcomes:

1. a point-in-time comparator universe and mapping policy separating direct competitors, sector proxies, broad-cap-weighted and equal-weighted indexes, small-cap proxies, and macro references;
2. provider, adjustment, currency, session, timezone, and completed-bar rules;
3. pair start dates, visible windows, horizon grids, fit/calibration/evaluation fractions, and missingness thresholds;
4. representation endpoints (support, stability, sensitivity), predictive endpoints, and separately cost-aware economic endpoints;
5. robust, exponentially weighted, and shrinkage beta comparators selected before holdout;
6. dependence-aware uncertainty and multiplicity correction across coordinates, pairs, and windows;
7. an untouched prospective holdout, a durable authenticated exposure ledger distinct from product analytics, and retained probe receipts; and
8. explicit promotion gates preserving current zero scanner and option-learning authority until incremental value and authority-boundary invariance are shown.

Property-based randomized identity/swap tests, suffix-mutation tests for pair context under a frozen split, expanded hash-tamper tests, and missing-provenance tests remain near-term software work. Runtime response-schema validation and a route probe are now implemented, but cold/concurrent distributions, browser time-to-first-useful-summary, interaction latency, pixel-diff baselines, manual screen-reader/color-vision review, and real-device checks remain open. Basket fields, point-in-time peer selection, cross-sectional ranking, and multivariate connectedness would be new model families rather than silent Pair-v1 extensions and should receive new version identifiers.

14 Reproducibility and Implementation Map

The numerical Pair-v1 operators were audited against implementation source 4c848dafecd41a64bd52844c9a12140e6534a784. The additive hierarchy, deterministic summary, exact window/support/compatibility disclosures, beta-chain metadata, and frozen-receipt contract were verified at 71e30b6b9168735816b02af93a82374c97a99ca3. The trust-language, reader-facing evidence, runtime, product-telemetry, probe, and browser-gate revision is bound to 0b8e8aa72be43cfd1038b983494f4693b50b4115. The later one-scope, chronological-trail, progressive-disclosure, and mobile hierarchy is bound to 55359cd301127fd771ea9fb2567195cf8d542688. The current goal-led paper source and release-gate closeout are identified by results/pdf_build_receipt.json. None changes a coordinate, relative-index, beta, field-separation, receipt, hash, or authority formula. The revision also avoids calling Equation 26 an OLS residual.

Artifact	Role
backend/app/services/market_weather_comparison.py	Pair construction, equations, masks, alignment, hashes, and authority receipt
backend/app/api/market_weather.py	Parameter validation, provider routing, cache key, and HTTP semantics
backend/tests/test_market_weather_comparison.py	Deterministic algebraic, chronology, precision, alignment, and cache tests
frontend/src/types/marketWeather.ts	Literal Pair-v1 response contract
frontend/src/components/marketWeather/MarketWeatherComparisonLab.tsx	Research visualization and translation layer
docs/papers/market-field/relative-field-addendum.tex	This source

Table 5: Source-level implementation map.

The focused backend tests generate their own synthetic inputs and do not require redistributable provider data. Exact live pair responses still depend on provider/cache history and cannot be reconstructed from this addendum alone. The component and pair hashes help compare identities when those inputs are available; they do not recover excluded data.

15 Conclusion

Relative Field Pair v1 is best understood as a typed, ordered comparison operator. Its strongest properties are explicitness: same-recipe legs, exact key intersection, bilateral support, two separate coordinate bases, full-precision price context, strictly prior beta estimation, fixed failure semantics, a self-checking unkeyed compact receipt, and zero authority. Those properties make it suitable for inspection and prospective research. They do not make the chosen benchmark correct, the pair stationary, the gap predictive, or the output economically useful. Those questions remain empirical and must be answered under a frozen, dependence-aware, cost-aware protocol before any promotion.

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A Exact Pair-v1 Constants

Constant	Value
Schema	market_field_pair_v1
Alignment contract	pair_alignment_v1
Normalization contract	native_and_fixed_proper_fit_relative_v1
Minimum exact shared observations	20
Maximum prior beta returns	60
Minimum prior beta returns	20
Minimum benchmark population return std.	10^{-7}
Maximum accepted absolute beta	25
Stored fit-scale validity floor	10^{-12}
Relationship comparison offset	5 common observations
Relationship absolute tolerance floor	0.05
Relationship relative tolerance	5% of prior stretch
Scanner and option-learning weights	0.0

B Response Field Interpretation

Response area	Interpretation
target, benchmark	Requested/canonical/provider identities, analysis hash, data source, latest aligned close, and latest returned close
window, support, compatibility	Requested/available/returned shared counts, exact bilateral cell support, and explicit session/currency/adjustment/timestamp disclosures
overlap	Visible common range, exact alignment rule, support fraction, dropout counts, unmatched tails, and session-compatibility status
coordinates	Metadata, latest values/support, and complete visible series for target, benchmark, native gap, and context gap
price_series	Full-precision-derived closes, relative index, active progress, prior beta, and cumulative beta-adjusted return
relative_progress	Latest aligned price, beta-chain metadata, and exact field-separation values/classification
summary	Deterministic non-predictive translation of authoritative result and support fields
frozen_receipt	Exact shared keys, latest evidence, identities, disclosures, zero authority, and self-checking unkeyed receipt hash
X-Market-Weather-Receipt-Hash	Response-header copy of the unkeyed frozen-receipt identity checksum
provenance	Component identities, ordered comparison hash, canonical/provider aliases, and contract versions
cache	Per-leg history lineage and per-request analysis-cache behavior
runtime	Request-local cache/build/response-ready timing and explicit post-return measurement exclusions; not part of either receipt identity
authority	Research-only mode and zero decision/execution authority
caveats	Reader-facing interpretation boundaries

C AI-Assistance Disclosure

OpenAI Codex materially assisted repository and formula inspection, claim-boundary review, test planning, LaTeX drafting, compilation checks, and consistency review. The human author remains responsible for the mathematics, citations, code, artifacts, and all empirical or operational claims. No language-model output is treated as empirical evidence or authorship.